



Chapter 4

Time Value of Money

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Time Value of Money

(6 hours)

4.1 Money (Type, functions and time value of money)

4.2 Simple and compound interests (Nominal, effective and continuous compounding)

4.3 Economic equivalence

4.4 Cash flow types (Single, uniform, linear gradient, geometric gradient and irregular)

Money (Type, functions and time value of money)

Money is anything that is generally accepted as a medium of exchange for goods and services, a measure of value, and a store of wealth in an economy.

In simple words, money is a tool that makes buying, selling, saving, and measuring value possible.

Types of Money

1. *Commodity Money*

Money that has intrinsic value.

Examples: Gold, silver, copper, salt

Used in ancient economies

2. *Metallic Money*

Money made of metals and issued in the form of coins.

Examples: Gold coins, silver coins

Durable but heavy and limited in supply

Money (Type, functions and time value of money)

3. *Paper Money (Fiat Money)*

Money made of paper and issued by the government or central bank.

Examples: Currency notes (NPR 100, 500, 1000)

Has no intrinsic value but legal backing

4. *Credit Money*

Money that is used based on trust and promise to pay.

Examples: Cheques, demand drafts, credit cards

5. *Bank Money*

Money deposited in banks and transferable through banking instruments.

Examples: Savings account, current account balances

Money (Type, functions and time value of money)

6. Plastic Money

Electronic form of money used for digital transactions.

Examples: Debit cards, credit cards, mobile banking, e-wallets

Money (Type, functions and time value of money)

I. Primary Functions

(a) Medium of Exchange

Money is used to buy and sell goods and services.

Example: Paying a contractor for construction work

(b) Measure of Value (Unit of Account)

Money expresses the value of goods and services in monetary terms.

Example: Cost of cement = NPR 1,200 per bag

2. Secondary Functions

(a) Store of Value

Money can be saved for future use.

Example: Saving salary for future construction projects

Money (Type, functions and time value of money)

(c) Transfer of Value

Money can be transferred from one person or place to another.

Example: Paying suppliers through bank transfer

3. Contingent Functions

(a) Basis of Credit

Money supports credit creation in the economy.

(b) Distribution of National Income

Used to distribute wages, rent, interest, and profit.

(c) Liquidity

Money is the most liquid asset.

Time Value of Money

The Time Value of Money (TVM) is a fundamental financial principle stating that money available today is worth more than the same amount in the future because of its potential earning capacity. This core concept explains why interest exists: it compensates lenders for forgoing the use of their money today.

The Time Value of Money (TVM) is influenced by several critical economic and financial factors that determine how much a specific sum of money is worth at different points in time.

I. Inflation

Inflation is the rate at which the general level of prices for goods and services rises, which subsequently erodes the purchasing power of money.

Purchasing Power: As inflation increases, a fixed amount of money buys fewer goods in the future than it does today.

Impact on Value: Higher expected inflation decreases the future value of money, making present cash flows significantly more desirable to avoid the loss of real value.

Time Value of Money

2. Interest Rates

Interest rates represent the "cost of money" or the reward for deferring current consumption.

Earning Potential: Higher interest rates increase the future value of an investment because the money grows faster over time.

Discounting: In present value calculations, a higher interest rate (discount rate) leads to a lower present value for future cash flows, reflecting a higher opportunity cost for waiting.

3. Risk and Uncertainty

Future cash flows are rarely guaranteed. Risk accounts for the possibility that the actual return on an investment will differ from the expected return.

Risk Premium: To compensate for uncertainty, investors require a higher rate of return on riskier investments. This effectively increases the discount rate, lowering the present value of those future sums.

Default Risk: The chance that a borrower may not be able to make promised payments also impacts the calculated value of future money.

Time Value of Money

4. Opportunity Cost

- Opportunity cost is the potential return lost by choosing one investment over another or by holding cash instead of investing it.
- Missed Gains: Money held today has the potential to earn interest immediately; every day it is not invested represents a missed opportunity for growth.
- Comparative Value: The value of receiving money in the future is always compared against what that money could have earned if it were available to be invested now.

5. Time Horizon

- The duration of the investment or loan period significantly changes the impact of TVM.
- Compounding Effect: The longer the time horizon, the more powerful the effect of compounding (earning interest on interest), leading to a much higher future value.
- Discounting Over Time: Conversely, the further into the future a payment is expected, the lower its present value becomes.

Time Value of Money

6. Preference for Consumption

- Most individuals naturally prefer current consumption over future consumption (time preference).
- Liquidity Preference: People generally value having liquid cash available now to meet immediate needs or take advantage of unexpected opportunities.
- Compensation for Waiting: To convince someone to give up current consumption, a "premium" in the form of interest must be paid.

Simple Interest

Simple interest is calculated exclusively on the original principal amount. It does not account for interest earned in previous periods, meaning the interest amount remains constant for every time cycle if the rate and principal do not change.

- **Formula:** $SI = P \times r \times t$
 - SI : Simple Interest
 - P : Principal amount (initial investment or loan)
 - r : Annual interest rate (expressed as a decimal)
 - t : Time period (usually in years)

Total Amount (A): ($A=P+SI$) or $A=P(1+rt)$

Common Uses: Used for short-term loans, auto loans, student loans, and some mortgages.

Compound Interest

Compound interest is calculated on the initial principal and also on the accumulated interest from previous periods. This "interest on interest" effect allows investments to grow at an accelerated rate over time.

- **Formula:** $A = P(1 + \frac{r}{n})^{nt}$
 - A : Future value of the investment/loan
 - n : Number of times interest is compounded per year

Common Uses: Typical for savings accounts, investment portfolios, and credit card balances.

Nominal Interest Rate

The nominal interest rate is the stated or advertised annual interest rate that does not account for the effects of compounding within that year.

Characteristics: It remains constant regardless of how often interest is compounded. However, it is not a complete measure of cost or return because it ignores compounding frequency.

Example: A credit card advertised at "18% interest per year, compounded monthly" has a nominal rate of 18%.

Effective Interest Rate: The effective interest rate (also known as the Effective Annual Rate or EAR) reflects the true annual cost of a loan or the true return on an investment by accounting for compounding frequency.

- **Formula:** $i_e = (1 + \frac{r}{n})^n - 1$
 - i_e : Effective annual interest rate
 - r : Nominal annual interest rate
 - n : Number of compounding periods per year

Simple Interest Compound Interest, Nominal Interest rate, Effective Interest rate

Example: A credit card company charges 21% interest per year, compounded monthly. What effective annual interest rate does the company charge?

$$r = 0.21 \text{ per year}$$

$$m = 12 \text{ months per year}$$

$$i_a = [1 + (.21 / 12)]^{12} - 1$$

$$= [1 + 0.0175]^{12} - 1$$

$$= (1.0175)^{12} - 1 = 1.2314 - 1$$

$$= 0.2314 = 23.14\%$$

Simple Interest Compound Interest, Nominal Interest rate, Effective Interest rate

The primary difference is that the nominal interest rate is the stated, annual rate that does not account for compounding, while the effective interest rate (also known as the Annual Equivalent Rate or AER) is the actual rate earned or paid, adjusted for the effect of compounding over the year.

Impact of Compounding: The more frequently interest is compounded (daily, monthly, quarterly), the higher the effective interest rate will be compared to the nominal rate (unless it compounds annually, in which case they are the same).

Formula: The effective annual rate (EAR) can be calculated from the nominal rate (r) and the number of compounding periods per year (n) using the formula: $EAR = (1 + r/n)^n - 1$.

Regulatory Disclosure: Regulators often require financial institutions to disclose both rates to ensure transparency and help consumers make informed decisions.

Simple Interest Compound Interest, Nominal Interest rate, Effective Interest rate

Nominal Interest Rate	Effective Interest Rate (EAR / AER)
The stated or advertised annual interest rate on a loan or investment.	The actual annual rate of interest earned or paid after considering compounding.
Does not account for the effect of compounding within the year.	Fully accounts for compounding (monthly, quarterly, semi-annual, etc.).
Calculated only on the original principal amount (similar to simple interest for one year).	Calculated on the principal plus accumulated interest (interest on interest).
Appears lower and is often used in advertisements to attract borrowers.	Gives a realistic picture of the true cost or return, helping better financial decisions.
Less accurate in representing the true financial impact.	More accurate and reliable for financial evaluation.
Difficult to compare loans/investments with different compounding frequencies.	Enables direct and fair comparison of financial products with different compounding periods.
Commonly used as an input rate but needs conversion before analysis.	Preferred for NPV, AW, FW, and benefit–cost analysis.

Continuous Compounding

Continuous compounding is a method of calculating interest where interest is added an infinite number of times per year.

In practice, it represents the maximum possible growth of money for a given interest rate.

- Long-term infrastructure investments
- Life-cycle cost analysis
- Projects with continuously changing cash flows

Continuous Compounding

Snowball Effect: Compound interest is often described as a snowball rolling downhill; as it rolls, it picks up more snow (interest), growing larger and faster with each rotation (compounding period).

Compounding Frequency: The more frequently interest is calculated and added to the balance (daily, monthly, quarterly, etc.), the faster the total amount grows. Daily or monthly compounding generally results in higher returns for savers than annual compounding, given the same nominal interest rate.

Time Value: The power of compounding becomes more significant over longer periods. Starting to save or invest early is a key strategy for maximizing the benefits of compound interest

Continuous Compounding

Continuous compounding is derived from the limit of discrete compounding:

$$\left(1 + \frac{r}{n}\right)^{nt} \quad \text{as } n \rightarrow \infty$$

This leads to the exponential function e .

Continuous Compounding

(a) Future Value (FV)

$$FV = PV e^{rt}$$

Where:

- PV = Present Value
- FV = Future Value
- r = annual interest rate (decimal)
- t = time in years
- $e = 2.71828$

Continuous Compounding

Example I: If \$100 is invested at 8% interest per year, compounded continuously, how much will be in the account after 5 years?

$$P = \$100$$

$$r = 8\%$$

$$n = 5 \text{ years}$$

$$F = P e^{r n} = (\$100) e^{(.08)(5)}$$

$$= (\$100) e^{0.4} = (\$100)(1.4918) = \$149.18$$

Continuous Compounding

Example 2: If \$100 is invested at 0.667% interest per month, compounded continuously, how much will be in the account after 5 years?

$$P = \$100$$

$$r = 0.666667\%$$

$$n = 5 \text{ years} * 12 \text{ months}$$

$$F = P e^{r n} = (\$100) e^{(.00666667)(60)}$$

$$= (\$100) e^{0.4} = (\$100)(1.4918) = \$149.18$$

Formulas of continuous compounding

Compounding Type	Periods per Year (m)	Future Value (FV) Formula
Annual	1	$FV = PV(1 + r)^t$
Semi-Annual	2	$FV = PV \left(1 + \frac{r}{2}\right)^{2t}$
Quarterly	4	$FV = PV \left(1 + \frac{r}{4}\right)^{4t}$
Continuous	∞	$FV = PVe^{rt}$

Questions

1) A contractor deposits NPR 1,000,000 in a bank at 10% nominal interest for 3 years.

Calculate the future value if interest is:

- a) compounded annually
- b) compounded quarterly

2) A building maintenance fund of NPR 2 million earns interest at 8% per annum, compounded semi-annually.

Find the amount after 5 years.

3) A road construction company takes a loan of NPR 5 million at 12% annual interest, compounded quarterly, for 2 years.

Calculate the total amount to be repaid.

Questions

- 4) A bridge will require NPR 30 million for rehabilitation after 12 years.
If the discount rate is 7% continuously compounded, find the present value.
- 5) A government plans to invest NPR 50 million in a water supply project for 15 years at 9% interest.
- a) Calculate the future value using:
- Annual compounding
 - Quarterly compounding
 - Continuous compounding
- 6) If a sum of Rs. 10,000 is deposited in a savings account at a nominal interest rate of 10% per annum, what will be the amount after 5 years if the interest is compounded:
- (i) Semi-annually
(ii) Monthly

Questions

7) If Rs. 5,000 is deposited every month for a period of 5 years, what will be the total amount accumulated at the end of 10 years, given that the nominal interest rate is 10% per annum, compounded quarterly?

8) A person invests a sum of NPR 50,000 in a bank at a nominal interest rate of 18% per annum, compounded monthly, for a period of 15 years. Find the maturity amount after 15 years.

Questions

7) If Rs. 5,000 is deposited every month for a period of 5 years, what will be the total amount accumulated at the end of 10 years, given that the nominal interest rate is 10% per annum, compounded quarterly?

- Future Value after 5 years of deposits (Annuity Calculation) Since the payment frequency (monthly) and the compounding frequency (quarterly) are different, we first need to find the equivalent interest rate per month.
- Nominal Annual Interest Rate (r): 10% or 0.10
- Compounding Frequency (m): 4 times per year (quarterly)
- Payments per Year (P/Y): 12 times per year (monthly)
- Given periodic interest rate (i) per compounding period: ($i=r/m=0.10/4=0.025$)

Equivalent monthly interest rate:

$$i_m = (1 + 0.025)^{4/12} - 1 \approx 0.00826484$$

Number of payments (n): 5 years * 12 payments/year = 60 payments

Monthly Payment (PMT or A): Rs. 5,000

$$FV_5 = A \left[\frac{(1 + i_m)^{60} - 1}{i_m} \right]$$

$$FV = 5000 \times \frac{(1 + 0.00826484)^{60} - 1}{0.00826484} \approx \text{Rs. } 3,89,754.21$$

Now, Grow this amount for the next 5 years (no deposits)

This approx. amount (Rs. 3,89,754.21) now grows with compound interest for another 5 years (from year 5 to year 10) with no further deposits.

- Principal (PV): Rs. 3,89,754.21
- Nominal Annual Interest Rate (r): 10% or 0.10
- Compounding Frequency (m): 4 times per year (quarterly)
- Time (t): 5 years
- Periodic interest rate (i): ($i=r/m=0.10/4=0.025$)
- Total number of compounding periods (n): $m \times t = 4 \times 5 = 20$

$$FV = PV \times (1 + i)^n$$

$$FV = 3,89,754.21 \times (1 + 0.025)^{20} \approx \text{Rs. } 3,89,754.21 \times 1.638616 \approx \text{Rs. } 6,38,582.49$$

Questions

Q) How long will it take the money to triple itself if invested at 10% compounded semi-annually?

we can use the compound interest formula:

$$A = P(1 + r/n)^{nt}$$

Where:

A = the amount of money after t years

P = the principal amount (initial investment)

r = annual interest rate (in decimal form)

n = number of times the interest is compounded per year

t = time in years

In this case, we want the money to triple, so $A = 3P$. The annual interest rate is 10% or 0.10, and it is compounded semi-annually, so $n = 2$.

Now, we substitute the values into the formula:

$$3P = P(1 + 0.10/2)^{2t}$$

Questions

Solving for t:

$$3 = (1 + 0.05)^{(2t)}$$

$$3 = (1.05)^{(2t)}$$

Taking the logarithm of both sides:

$$\ln(3) = \ln(1.05)^{(2t)}$$

$$\ln(3) = 2t * \ln(1.05)$$

$$t = \ln(3) / (2 * \ln(1.05))$$

Calculating the value:

$$t \approx \ln(3) / (2 * \ln(1.05))$$

$$t \approx 11.9 \text{ years}$$

Questions

Q) Consider a situation in which money flows daily. Suppose you own a retail shop and generate Rs 200 cash each day. You establish a special business account and deposit your daily cash flows in an account for 15 months. The account earns an interest rate of 6%. Compare the accumulated cash values at the end of 15 months, assuming (a) Daily compounding

(a) Daily compounding.

Given: $A = \text{Rs } 200$ per day, $r = 6\%$ per year, $M = 365$ compounding periods per year, and $N = 455$ days.

Find: F .

Assuming that there are 455 days in the 15-month period, we find that

$$i = 6\%/365$$

$$= 0.01644\% \text{ per day}$$

$$N = 455 \text{ days}$$

The balance at the end of 15 months will be

$$F = \text{Rs } 200(F/A, 0.01644\%, 455)$$

$$= \text{Rs } 200(472.4095)$$

$$= \text{Rs } 94,482$$

Economic Equivalence

Interpreting the Economic Equivalence

Interpreting economic equivalence means understanding that despite differences in magnitude and timing, various financial transactions can hold the same economic weight under a given interest rate. For example, receiving \$100 today might be economically equivalent to receiving \$105 one year from now if the prevailing interest rate is 5%. This equivalence implies indifference between the two options from a purely economic standpoint. When evaluating projects or investments, financial analysts apply this principle by converting all projected cash flows to a single reference point, typically the present, using a consistent discount rate. This allows for a direct comparison of alternatives, facilitating optimal financial analysis and resource allocation. If the values are equivalent, an investor would theoretically be indifferent between the alternatives; if not, one option would be preferable.

Economic Equivalence

Hypothetical Example:

Consider a scenario where an individual, Sarah, has two investment options for a sum of money:

Option A: Receive \$10,000 today.

Option B: Receive \$11,500 three years from now.

To determine which option offers greater economic equivalence, Sarah needs to choose a relevant interest rate to serve as her opportunity cost. Let's assume her alternative investment opportunity (e.g., a low-risk bond) yields 4.5% annual compounding.

To compare these, Sarah can calculate the present value of Option B:

Economic Equivalence

$$\begin{aligned}PV &= \frac{FV}{(1+i)^n} \\PV &= \frac{\$11,500}{(1+0.045)^3} \\PV &= \frac{\$11,500}{(1.045)^3} \\PV &= \frac{\$11,500}{1.141166} \\PV &\approx \$10,077.30\end{aligned}$$

In this example, the present value of receiving \$11,500 in three years is approximately \$10,077.30 when discounted at 4.5% annually. Since \$10,077.30 (Option B's present value) is greater than \$10,000 (Option A's present value), Option B is the economically preferable choice, as it yields a slightly higher equivalent value today

Economic Equivalence

Practical Applications

Economic equivalence is a cornerstone in numerous real-world financial contexts. In capital budgeting, businesses use it to evaluate long-term investment projects, comparing the present value of anticipated cash inflows against initial costs to decide project viability. This is commonly done through methods like Net Present Value (NPV) and Discounted Cash Flow (DCF) analysis. For instance, when considering the purchase of new machinery versus leasing it, economic equivalence helps determine which option has a lower total cost when adjusted for the time value of money. Financial institutions employ this principle to price bonds, loans, and other financial instruments, ensuring that the stream of future payments or receipts is equivalent to the instrument's current market price. Corporate Finance Institute outlines how DCF analysis is used to evaluate investments by discounting estimated future cash flows to their present value. Retirement planning also heavily relies on economic equivalence, as individuals project future financial needs and determine the present-day contributions required to meet those goals, considering factors like inflation.

Development of Interest Formulas

Table 1: Basic Payments & Uniform Series

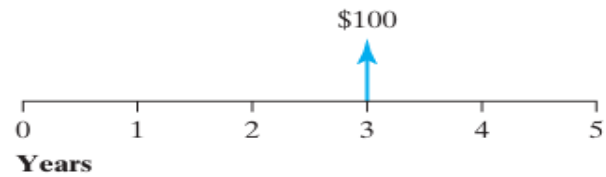
Concept	Formula
Single Payment (FV)	$F = P(1 + i)^n$
Single Payment (PV)	$P = F(1 + i)^{-n}$
Uniform Series (FV)	$F = A \frac{(1+i)^n - 1}{i}$
Uniform Series (PV)	$P = A \frac{(1+i)^n - 1}{i(1+i)^n}$

Development of Interest Formulas

Table 2: Gradients & Special Cases

Concept	Formula
Capital Recovery (A from P)	$A = P \frac{i(1+i)^n}{(1+i)^n - 1}$
Arithmetic Gradient (FV)	$F = G \frac{(1+i)^n - in - 1}{i^2}$
Geometric Gradient (PV)	$P = A_1 \frac{1 - \left(\frac{1+g}{1+i}\right)^n}{i-g}, g \neq i$
Continuous Compounding (FV)	$F = Pe^{in}$

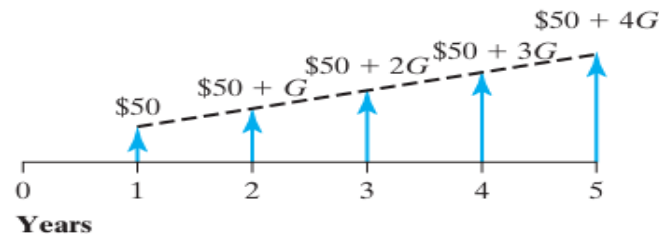
Cash flow types (Single, uniform, linear gradient, geometric gradient and irregular)



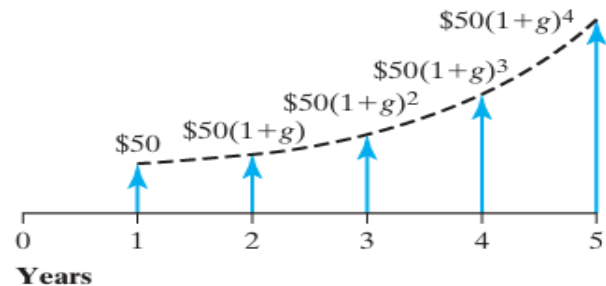
(a) Single cash flow



(b) Equal (uniform) payment series at regular intervals



(c) Linear-gradient series, where each cash flow in the series increases or decreases by a fixed amount G



(d) Geometric-gradient series, where each cash flow in the series increases or decreases by a fixed rate (percentage) g

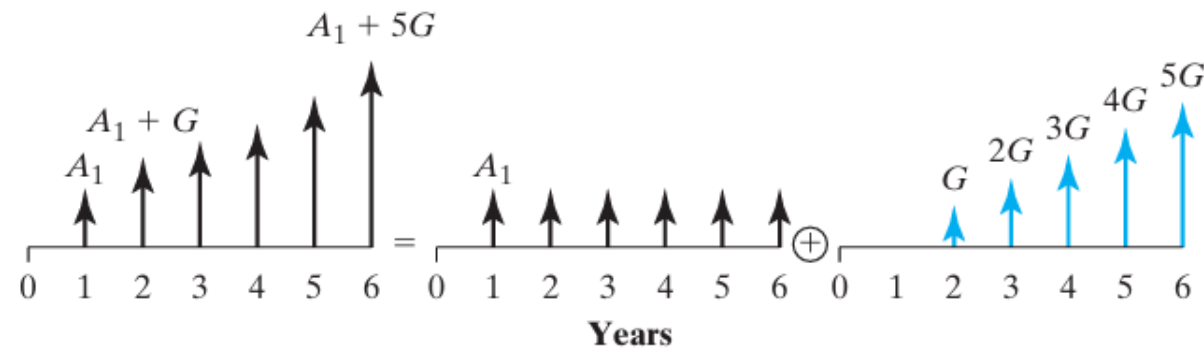


(e) Irregular payment series, which exhibits no regular overall pattern

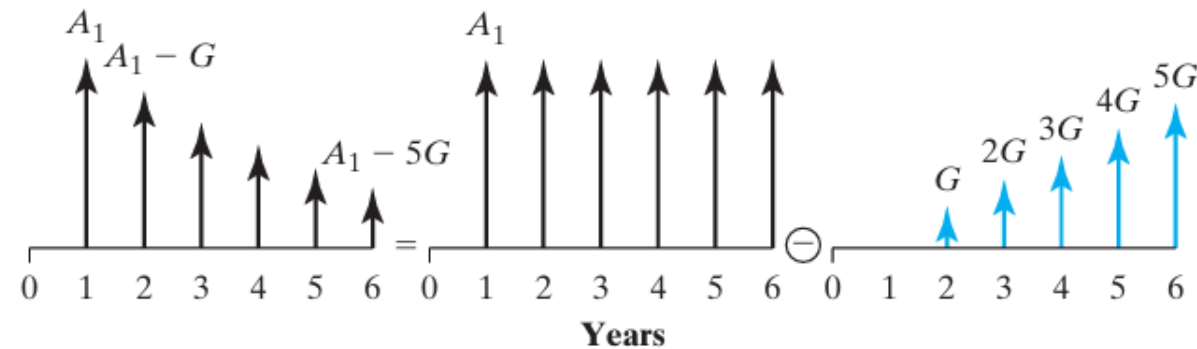
Five types of cash flows:

- (a) Single cash flows,
- (b) equal (uniform) payment series,
- (c) linear-gradient series,
- (d) geometric-gradient series, and
- (e) irregular payment series.

Cash flow types (Single, uniform, linear gradient, geometric gradient and irregular)



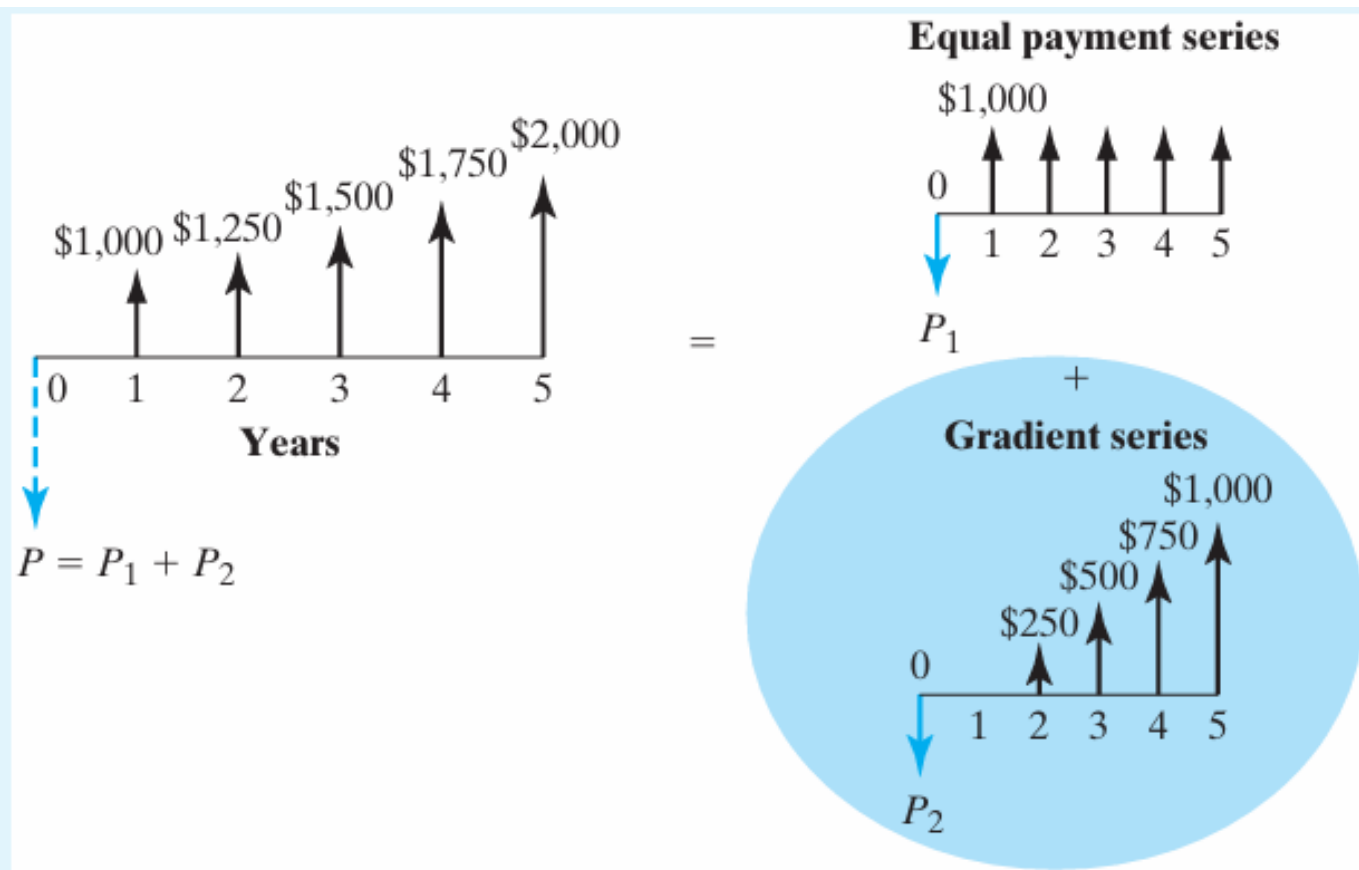
(a) Increasing gradient series



(b) Decreasing gradient series

Figure 3.3 I Two types of linear-gradient series as composites of a uniform series of N payments of A_1 and the gradient series of increments of constant amount G .

Cash flow types (Single, uniform, linear gradient, geometric gradient and irregular)



A. The first component is an equal-payment series (AI), and the second is a linear-gradient series (G). We have

$$P = P_1 + P_2$$

$$P = AI(P/A, 12\%, 5) + G(P/G, 12\%, 5)$$

$$= \$1,000(3.6048) + \$250(6.397)$$

$$= \$5,204$$

Cash flow types (Single, uniform, linear gradient, geometric gradient and irregular)

Gradient-to-Equal-Payment Series Conversion Factor: Find A, Given G, i, and N

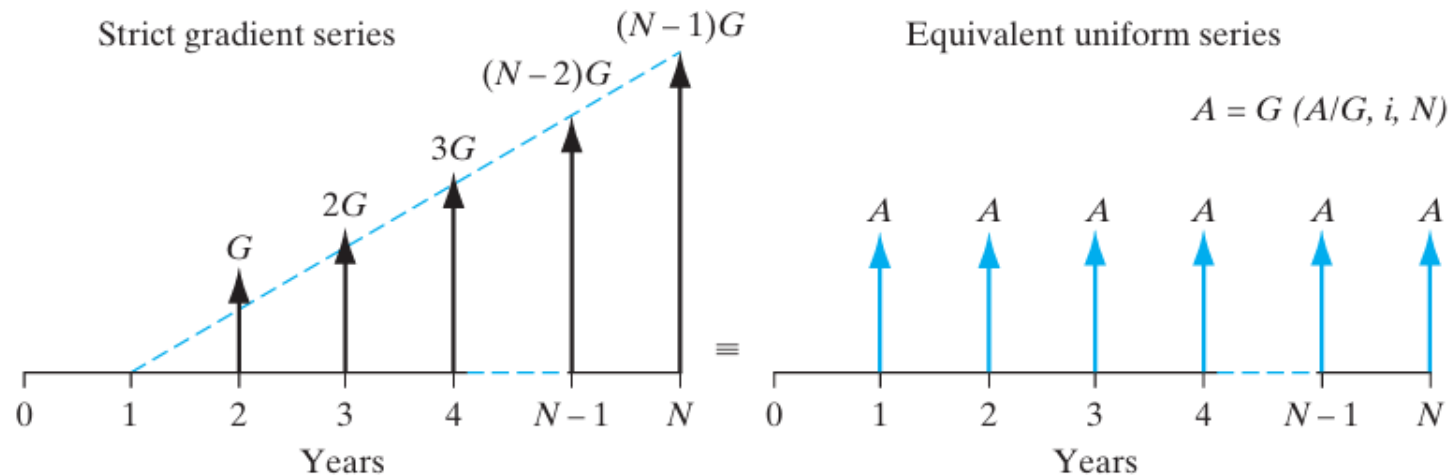
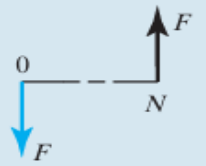
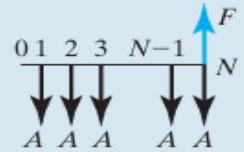
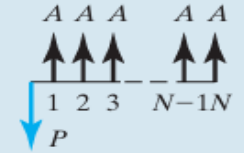


Figure 3.34 Conversion of a gradient series to its equivalent uniform series.

$$A = G \left[\frac{(1 + i)^N - iN - 1}{i[(1 + i)^N - 1]} \right] = G(A/G, i, N)$$

Flow Type	Factor Notation	Formula	Excel Command	Cash Flow Diagram
S I N G L E	Compound Amount ($F/P, i, N$) Present Worth ($P/F, i, N$)	$F = P(1 + i)^N$ $P = F(1 + i)^{-N}$	$= \text{FV}(i, N, 0, P)$ $= \text{PV}(i, N, 0, F)$	
E Q U A L	Compound Amount ($F/A, i, N$)	$F = A \left[\frac{(1 + i)^N - 1}{i} \right]$	$= \text{FV}(i, N, A)$	
P A Y M E N T	Sinking fund ($A/F, i, N$)	$A = F \left[\frac{i}{(1 + i)^N - 1} \right]$	$= \text{PMT}(i, N, 0, F)$	
S E R I E S	Present worth ($P/A, i, N$) Capital recovery ($A/P, i, N$)	$P = A \left[\frac{(1 + i)^N - 1}{i(1 + i)^N} \right]$ $A = P \left[\frac{i(1 + i)^N}{(1 + i)^N - 1} \right]$	$= \text{PV}(i, N, A)$ $= \text{PMT}(i, N, P)$	

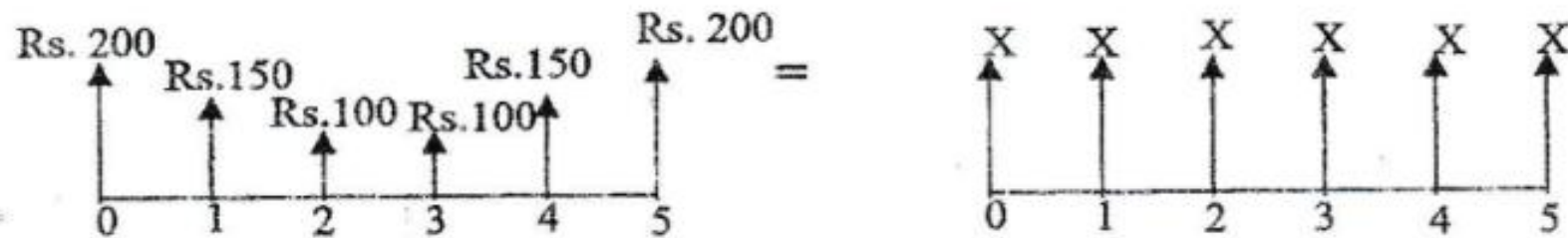
G R A D I E N T	Linear gradient		
	Present worth ($P/G, i, N$)	$P = G \left[\frac{(1 + i)^N - iN - 1}{i^2(1 + i)^N} \right]$	
	Equal-payment Conversion factor ($A/G, i, N$)	$A = G \left[\frac{(1 + i)^N - iN - 1}{i[(1 + i)^N - 1]} \right]$	
	Geometric gradient		
S E R I E S	Present worth ($P/A_1, g, i, N$)	$P = \left[\begin{array}{l} A_1 \left[\frac{1 - (1 + g)^N(1 + i)^{-N}}{i - g} \right] \\ A_1 \left(\frac{N}{1 + i} \right) \text{ (if } i = g \text{)} \end{array} \right]$	

Questions

1. A bridge will require a repair costing Rs 50,000 in 5 years. What is its present worth at 8% interest?
2. A road maintenance program requires Rs 10,000 annually for 7 years. Calculate the present worth at 6% interest.
3. A construction project requires Rs 20,000 now, Rs 10,000 after 2 years, and Rs 15,000 after 5 years. Find the total present worth at 10% interest.
4. An equipment will cost Rs 80,000 in 3 years. Determine its present worth at 9% interest.
5. A water supply system requires Rs 25,000 for maintenance after 1 year, Rs 30,000 after 3 years, and Rs 35,000 after 5 years. Find present worth at 7% interest.

Questions

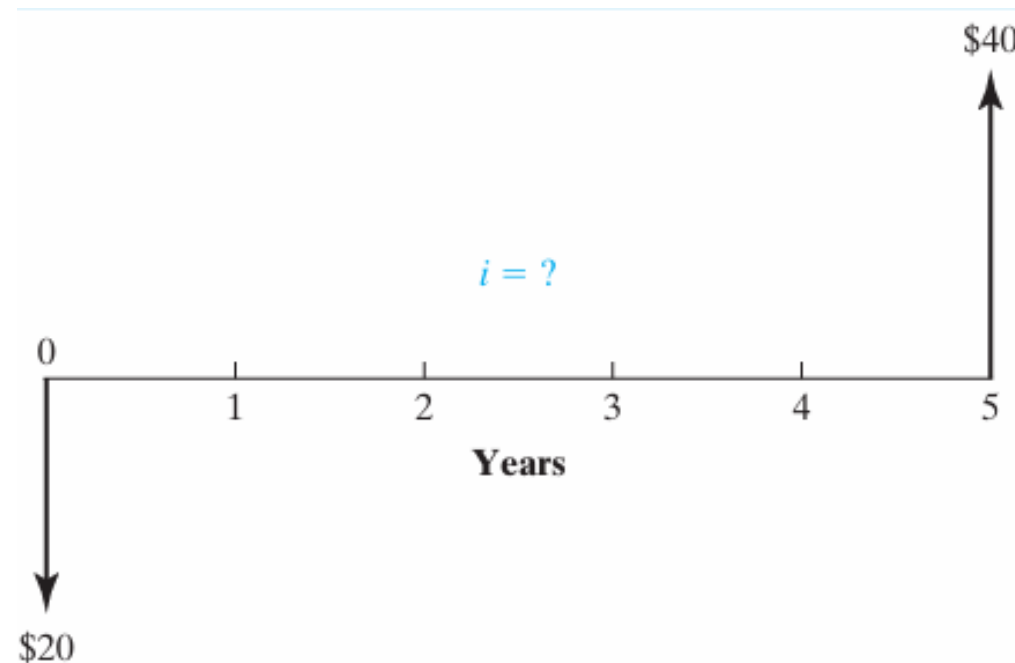
6. The two cash flow transactions shown below are said to be equivalent at 10% interest, compounded annually. Find the unknown X value which satisfies the equivalence.



7. Assume you contribute Rs. 5,000 monthly to a retirement account that earns an interest rate of 10%, compounded quarterly. Calculate the total balance after 12 years, considering that interest is applied to the deposits made during each compounding period.
8. Suppose that \$1,000 is to be received in five years. At an annual interest rate of 12%, what is the present worth of this amount?

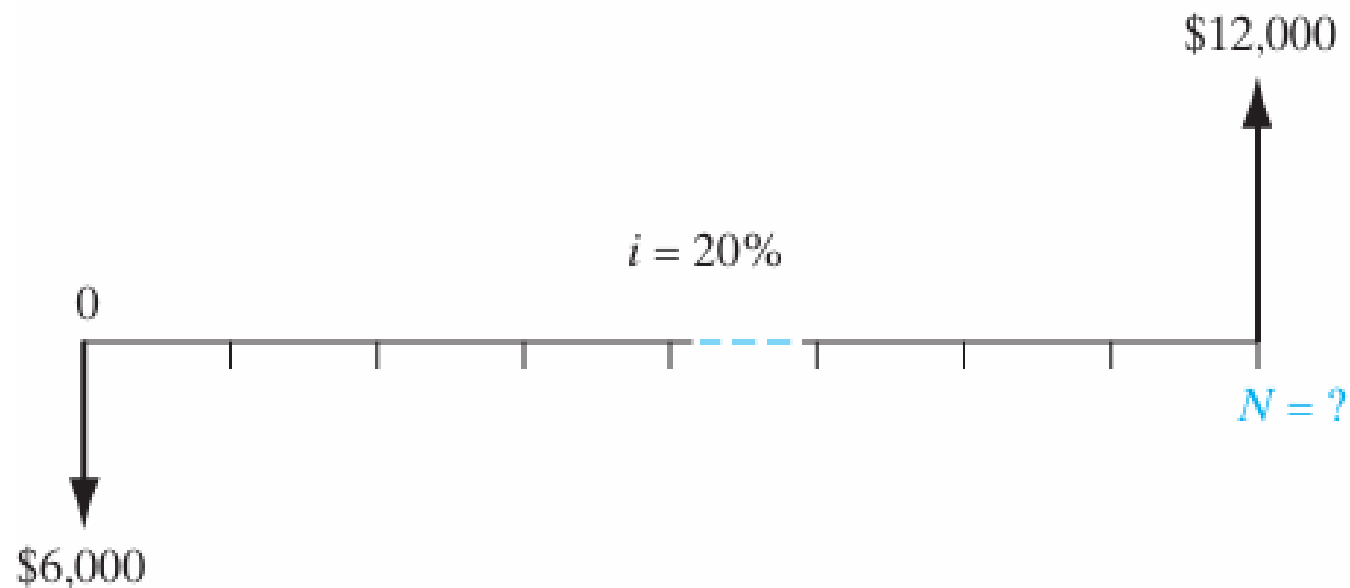
Questions

9) Suppose you buy a share of stock for \$20 and sell it for \$40. Then your profit is \$20. If that happens within a year, your rate of return (interest rate earned on your investment) is an impressive 100%. If it takes five years, what would be the compound annual rate of return on your investment?



Questions

10) You have just purchased 100 shares of Facebook stock at \$60 per share. You will sell the stock when its market price has doubled. If you expect the stock price to increase 20% per year, how long do you anticipate waiting before selling the stock



Questions

Gradient Series Questions

1. Annual maintenance cost of a road increases by Rs 500 each year, starting from Rs 2,000. Find present worth over 6 years at 10% interest.
2. Toll revenue increases by 5% annually starting from Rs 10,000. Find present worth over 8 years at 8% interest.
3. A pavement maintenance cost grows geometrically at 7% per year starting from Rs 4,000. Find equivalent annual worth over 10 years.
4. Equipment maintenance starts at Rs 1,500 per year and increases by Rs 200 annually for 5 years. Calculate present worth at 9% interest.
5. A bridge toll increases by Rs 1,000 every year for 7 years, starting from Rs 5,000. Determine annual worth at 6% interest.

Questions

6. An investment of \$30,000 today will accumulate over 10 years at 7% interest. Find its future value
7. A toll plaza collects \$5,000 annually for 8 years. Determine the future worth at 5% interest.
8. A pavement maintenance program costs \$8,000 per year for 6 years. Calculate the future worth at 6% interest.
9. Equipment costing \$40,000 today will generate \$6,000 per year for 7 years. Find its future worth at 8% interest.
10. A bridge replacement project requires annual payments of \$12,000 for 10 years. Determine future worth at 9% interest.

Questions

11. A road construction project costs \$200,000 with maintenance of \$10,000 per year for 15 years. Find annual worth at 6% interest.
12. Equipment costs \$50,000 with a salvage value of \$10,000 after 5 years. Find annual cost at 10% interest.
13. Bio-Gen Company, a small biotechnology firm, has borrowed \$250,000 to purchase laboratory equipment for gene splicing. The loan carries an interest rate of 8% per year and is to be repaid in equal installments over the next six years. Compute the amount of the annual installments

Questions

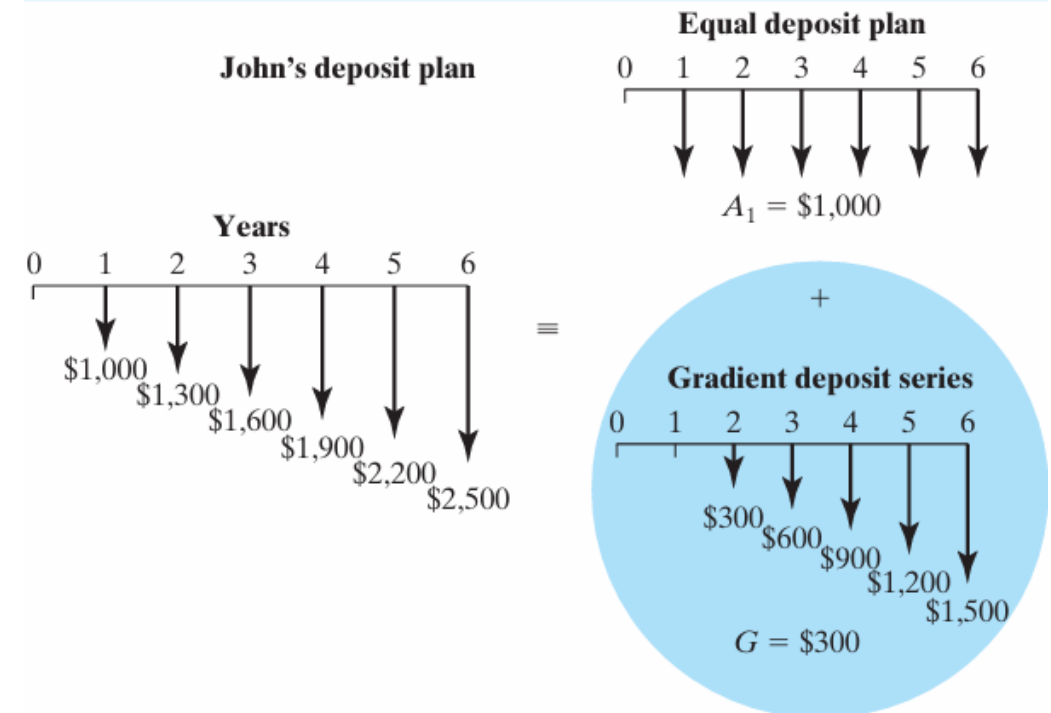
14) A textile mill has just purchased a lift truck that has a useful life of five years. The engineer estimates that maintenance costs for the truck during the first year will be \$1,000. As the truck ages, maintenance costs are expected to increase at a rate of \$250 per year over the remaining life. Assume that the maintenance costs occur at the end of each year. The firm wants to set up a maintenance account that earns 12% annual interest. All future maintenance expenses will be paid out of this account. How much does the firm have to deposit in the account now?

$A1 = \$1,000$, $G = \$250$, $i = 12\%$ per year, and $N = 5$ years. Find: P

Questions

15) John and Barbara have just opened two savings accounts at their credit union. The accounts earn 10% annual interest. John wants to deposit \$1,000 in his account at the end of the first year and increase this amount by \$300 for each of the next five years. Barbara wants to deposit an equal amount each year for the next six years. What should be the size of Barbara's annual deposit so that the two accounts will have equal balances at the end of six years

Given: $A_1 = \$1,000$, $G = \$300$, $i = 10\%$, and $N = 6$. Find: A .



Questions

➤ To find the equal payment series beginning at the end of year 1 and ending at year 6 that would have the same present worth as that of the gradient series, we may

proceed as follows:

$$A = \$1,000 + \$300(A/G, 10\%, 6)$$

$$= \$1,000 + \$300(2.2236)$$

$$= \$1,667.08$$

Barbara's annual contribution should be \$1,667.08

Additional important Questions

- I. A bank offers a nominal interest rate of 9% per annum compounded quarterly. Calculate the effective interest rate:
 - a) Per year
 - b) Per quarter
 - c) Per month
 - d) Per half-year

- I. Calculate the future worth at the end of 10 years at 12% interest compounded monthly of a cash flow of Rs. 40,000 deposited at the beginning of each year for 5 years.

2. Evaluate the future worth at the end of 15 years with 10% interest compounded monthly for a cash flow of Rs. 50,000 deposited at the beginning of each year for the first 10 years.

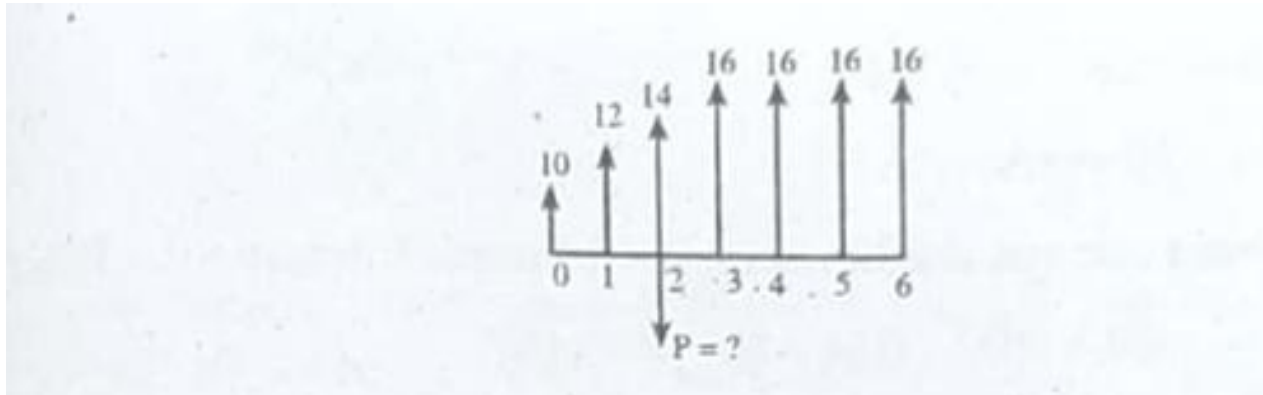
3. You deposit Rs. 6,000 per month for 3 years at 7% interest per year. Find the future sum at the end of 5th year.

4. Monthly deposits of Rs. 5,000 are made for 5 years at 8% interest compounded weekly. The interest rate then changes to 6% per year. Determine the total accumulated amount after 8 years.

5. How much amount must be deposited now so that Rs. 10,000 can be withdrawn per month for:
 - a) 50 years
 - b) ForeverAssume interest rate = 17% per annum.

6. An amount of Rs. 3,00,000 is borrowed at 14% interest compounded semi-annually and is to be repaid in equal annual installments over 10 years. Determine the annual installment
7. A person deposits Rs. 25,000 at the end of each year for 12 years in a bank that pays 9% interest compounded annually. Determine:
 - a) The total amount accumulated at the end of 12 years
 - b) The equivalent uniform annual deposit if the final amount is withdrawn over 6 years at the same interest rate amount.
8. Find the present worth of a deferred annuity of Rs. 15,000 per year payable for 8 years, if the first payment occurs at the end of the 5th year. Assume interest rate = 10% per annum.

9. Find the value of the present worth P if the interest rate is $i=10\%$ per annum. Use the gradient formula



10. A uniform series of Rs. 20,000 per year continues for infinite years. If the interest rate is 8% per year, determine the capitalized value of the series.

Thank you for your attention!